IMT 526 LLM

SOCIAL MEDIA TOXCITIY DETECTION

GROUP NAME:
Anti-Haters (Group 4)

PRESENTED BY:
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AGENDA

- INTRODUCTION
- PROBLEM STATEMENT
- DATASET & METHODOLOGY
- EVALUATION PLAN
- RESULTS & ANALYSIS
- LIMITATIONS & FUTURE WORK

PROBLEM

Social Media Toxcitiy Detection

510,000 *COMMENTS POSTED EVERY MINUTE*

MARR, B. (2021, JULY 2). HOW MUCH DATA DO WE CREATE EVERY DAY? THE MIND-BLOWING STATS EVERYONE SHOULD READ. BERNARD MARR. YOUR PARAGRAPH TEXT

Notes	Gen Z Slangs	Millennial
Means something is excellent (1)	Bussin'	The Bomb / That's Fire
Refers to charisma or charm (1)	Rizz	Swag
Means "no lie" or "for real" (1)	No Cap	Real Talk
Used to describe a vibe or feeling (1)	It's Giving	That's Lit / I Can't Even
Calling something questionable (1)	Side Eye/Sus	That's Suspicious
Refers to showing off an outfit (1)	Fit Check	OOTD (Outfit Of The Day)
Used for emphasis or sincerity (1)	On God	Ferreal
Affirmation of truth (1)	Word/Bible	That's Legit

INTERNATIONAL CENTER FOR LANGUAGE STUDIES. (2025, MAY 14). 8 FIRE GEN Z SLANG TERMS AND THE MILLENNIAL EQUIVALENT. ICLS. HTTPS://WWW.ICLS.EDU/BLOG/8-FIRE-GEN-Z-SLANG-TERMS-AND-THE-MILLENNIAL-EQUIVALENT

PROBLEM MOTIVATION

Why not off-the-shelf APIs?

- Existing toxicity APIs rely on encoder models trained on fixed datasets
- Can struggle with out-of-distribution language not seen during training
- Miss evolving language like slang, coded language, and deliberate misspellings

1. COMMENTS CAN APPEAR NON-TOXIC WHEN VIEWED IN ISOLATION BUT BECOME HARMFUL WHEN CONTEXT IS CONSIDERED— SONG ET AL., 2026

2. LLMS ARE CAPABLE OF RECOGNIZING TOXIC LANGUAGE AND CAPTURE CONTEXTUAL INFORMATION THAT TRADITIONAL TOXICITY METRICS OFTEN MISS — KOH ET AL. (2024)

DATA SET & EVALUATION METRICS

ORIGINAL DATASET

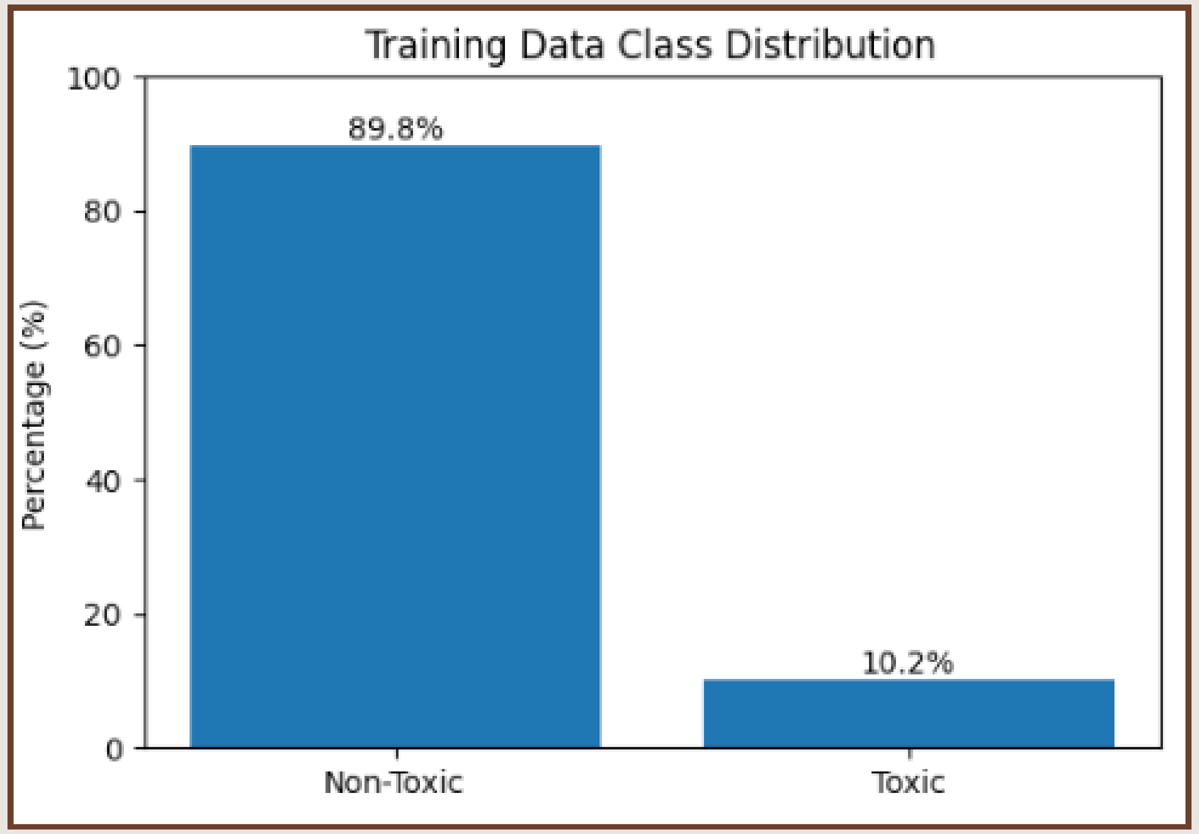
- Created by Jigsaw & Google for toxic content moderation
- ~**160k** labeled **training** set + ~**64k** labeled **test** samples

TOXIC | SEVERE TOXIC | OBSCENE | THREAT | INSULT | IDENTITY HATE

OUR DATASET

We simplify the task into **binary classification**: whether a comment belongs to one of the six toxic categories

TOXIC | NON-TOXIC



Due to imblanced dataset, we eliminate accuracy as our evaluation

METRICS

MACRO F1
Balances performance across classes

TOXIC F1
Toxic-class detection performance

PRECISION
Measures false positive control

TOXIC RECALL
Measures missed toxic comments

- We emphasize **toxic recall** since **missing toxic comments can be costly in real-world moderation systems**
- Different evaluation metrics capture different moderation risks.

METHODOLOGY

1 TF-IDF + LOGISTIC REGRESSION

TF-IDF

Convert text into TF-IDF features

TUNE USING VALIDATION DATA

- C (regularization strength): **10.0**
- ngram_range: **(1, 2)**

LOGISTIC REGRESSION

Predict the test dataset

2 LLAMA 3.1 8B WITH PROMPT

ZERO-SHOT PROMPT

" Classify the following text as toxic or non-toxic.

Desired Format: a number, 1 = toxic, 0 = non-toxic "

VAL DATA PREDICTION

Adjust prompt based on validation performance

FINAL TEST EVALUATION

3 LLAMA 3.1 8B QLORA FINE-TUNING

TRAINING SETUP

- Continue using zero-shot prompt
- Balanced training sample (1k toxic + 1k no-toxic)
- Tune learning rate and epochs using validation loss

QLORA TRAINING

- 4-bit quantization
- Low-rank adapters (LoRA)
- Imbalanced validation data for realistic evaluation

FINAL TEST EVALUATION

RESULTS & ANALYSIS

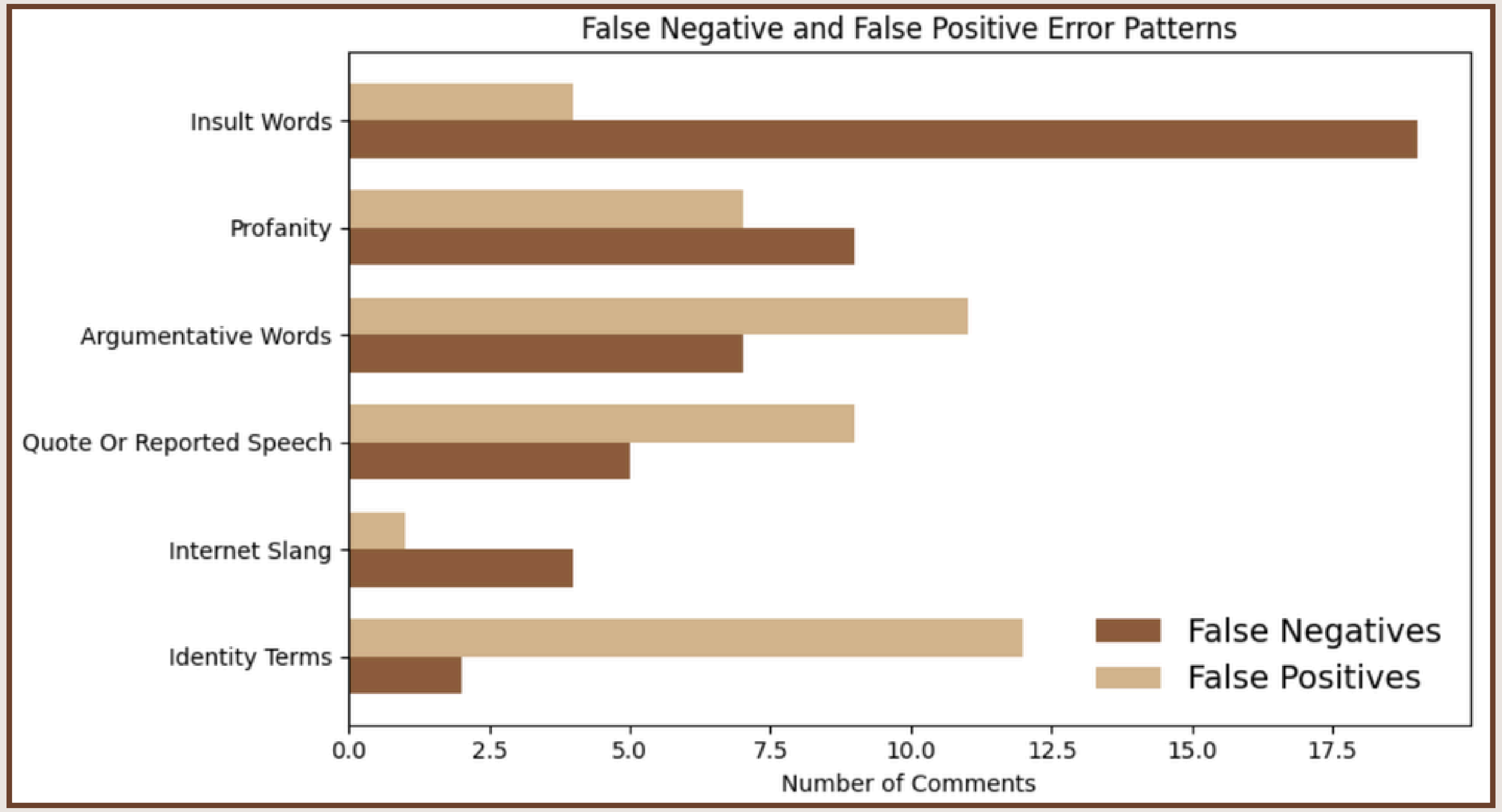
METRICS EVALUATION

Model	Macro F1	Toxic F1	Toxic Precision	Toxic Recall
TF-IDF + Logistic Regression	**0.778**	0.618	0.478	0.874
Zero-shot (llama 3.1 8B with prompt)	0.696	0.49	0.352	0.809
Qlora fine-tuning (llama 3.1 8B)	0.707	0.514	0.361	**0.88**

KEY FINDINGS

- **TF-IDF** ACHIEVED **THE STRONGEST OVERALL** CLASSIFICATION PERFORMANCE
- **QLORA** ACHIEVED THE **HIGHEST TOXIC RECALL**
- **FINE-TUNING** IMPROVED **SENSITIVITY TO SUBTLE TOXIC LANGUAGE** BUT INCREASED FALSE POSITIVES

PATTERNS ANALYSIS



FALSE NEGATIVES

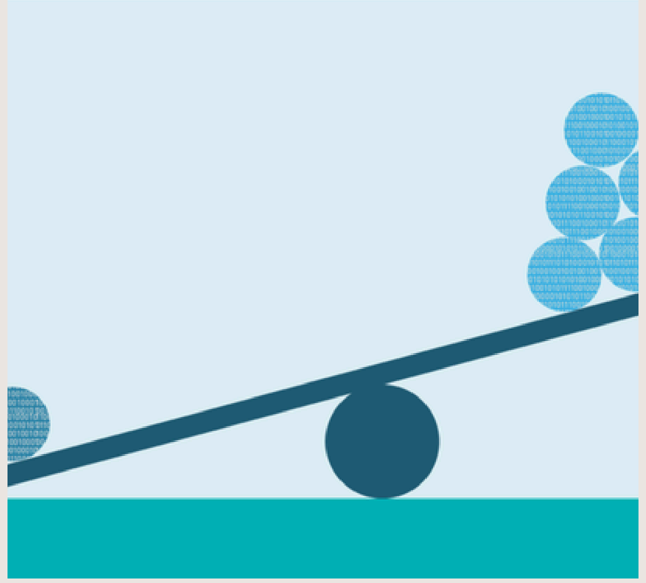
- MILD INSULTS
- SUBTLE CONVERSATIONAL TOXICITY
- CONTEXT-DEPENDENT HOSTILITY

FALSE POSITIVES

- EMOTIONALLY INTENSE DISCUSSIONS
- QUOTED TOXIC EXPRESSIONS
- LONG ARGUMENTATIVE COMMENTS

CONTEXTUAL TOXICITY REMAINS DIFFICULT TO DETECT

LIMITATIONS

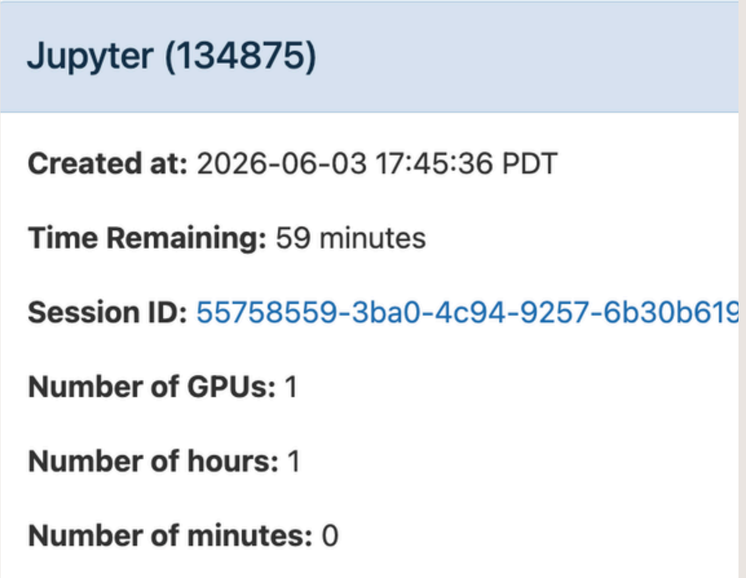


CLASS IMBALANCE

90.4% non-toxic comments, around 9.6% toxic comments

CONVERSATIONAL CONTEXT

Texts are evaluated individually



GPU HOURS

One hour Max and if more then long queue

FUTURE WORK

- Multi-label Classification — Extend the model to predict toxicity categories simultaneously
- Multilingual Support — Test and adapt the model for non-English comments
- Human-in-the-Loop Workflow — Use model confidence scores to help moderators prioritize flagged comments for review

Thank you